Chapter 14

Compund Interest

As we know that when we borrow some money from bank or any person, then we have to pay some extra money at the time of repaying. This extra money is known as **interest**. If interest accrued on principal, it is known as simple interest. Sometimes it happens that we repay the borrow money some late. After the completion of specific period, interest accrued on principal as well as interest due of the principal. Then, it is known as compound interest.

Compound interest = Amount - Principal

Basic Formulae Related to Compound Interest

Let principal = P, rate = R% pa and time = n yr

1. If interest is compounded annually, then

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^n$$

$$\text{Compound Interest} = \text{Amount - Principal}$$

$$\text{Compound Interest} = P\left[\left(1+\frac{R}{100}\right)-1\right]^n$$

2. If interest is compounded half-yearly, then $R=\frac{R}{2}$ and $n=2n$

$$\text{Amount} = P\left(1+\frac{R}{2\times 100}\right)^{2n}$$

3. If interest is compounded quaterly, then $R=\frac{R}{4}$ and $n=4n$

$$\text{Amount} = P\left(1+\frac{R}{4\times 100}\right)^{4n}$$

4. If interest is compounded annually but time is in fraction (suppose time $=n\frac{a}{b}$ yr),then

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^n \times \left(1+\frac{(a/b)R}{100}\right)$$

5. If rates of interest are $R_1\%$, $R_2\%$ and $R_3\%$ for 1st, 2nd and 3rd year respectively, then

$$\text{Amount } = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right)$$

Ex. 1 Find the compound interest on ₹8000 at 4% pa for 2 yr, compounded annually.

Sol. Given that, $P = ₹8000$, $R = 4\%$ and time, $n = 2\ yr$

Now, according to the formula,

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^n$$

$$= 8000\left(1+\frac{4}{100}\right)^2 = 8000 \times \frac{26}{25} \times \frac{26}{25} = ₹8652.80$$

$\therefore$ CI = Amount - Principal $= 8652.80 - 8000 = ₹652.80$

Ex. 2 Ruchi invested ₹1600 at the rate of compound interest for 2 yr. She got ₹1764 after the specified period. Find the rate of interest.

Sol. Given that, $P = ₹1600$, $n = 2\ yr$ and $A = ₹1764$

Now, according to the formula,

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^n \implies 1764 = 1600\left(1+\frac{R}{100}\right)^2$$

$$\implies \frac{1764}{1600} = \left(\frac{100+R}{100}\right)^2 \implies \frac{441}{400} = \left(\frac{100+R}{100}\right)^2$$

$$\implies \left(\frac{21}{20}\right)^2 = \left(\frac{100+R}{100}\right)^2 \implies \left(\frac{100+R}{100}\right) = \left(\frac{21}{20}\right)$$

$$\implies 100+R = \frac{21}{20} \times 100 \implies 100+R = 105$$

$\therefore$ $R = 105 - 100 = 5\%$

Ex. 3 Find the compound interest on ? 5000 in 2 yr at 4% pa, if the interest being compounded half-yearly.

Sol. Given that, $P = ₹5000$, $R = 4\%$ and time, $n = 2\ yr$

Now, according to the formula,

$$\text{Amount} = P\left(1+\frac{R}{2\times 100}\right)^{2n}$$

$$= 5000\left(1+\frac{4}{200}\right)^{4} = 5000\times\left(\frac{51}{50}\right)^{4}$$

$$= \frac{5000\times 51\times 51\times 51\times 51}{50\times 50\times 50\times 50} = ₹5412.16$$

$\therefore$ CI = Amount - Principal = $5412.16 - 5000 = ₹412.16$

Ex. 4 Find the compound interest on ₹8000 at 20% pa for 9 months, compounded quarterly.

Sol. Given that, $P = ₹8000,\ R = 20\%$ and $n = 9$ months

According to the formula,

$$\text{Amount} = P\left(1+\frac{R}{4\times 100}\right)^{4n}$$

$$= 8000\left(1+\frac{20}{400}\right)^{4\times\frac{9}{12}} = 8000\left(1+\frac{20}{400}\right)^{3}$$

$$= 8000\times\frac{21}{20}\times\frac{21}{20}\times\frac{21}{20} = ₹\ 9261$$

$\therefore$ CI = Amount - Principal = $9261 - 8000 = ₹\ 1261$

Ex. 5 Find the compound interest on ₹2000 at 15% pa for 2 yr 4 months, compounded annually.

Sol. Given that, $P = ₹\ 2000,\ R = 15\%,\ n = 9$ and $\frac{a}{b} = \frac{4}{12} = \frac{1}{3}$

According to the formula,

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^{n}\times\left(1+\frac{(a/b)R}{100}\right)$$

$$= 2000\left(1+\frac{15}{100}\right)^{2}\times\left(1+\frac{\frac{1}{3}\times 15}{100}\right)$$

$$= 2000\times\frac{23}{20}\times\frac{23}{20}\times\frac{21}{20} = \frac{11109}{4} = ₹\ 2777.25$$

$\therefore$ CI = Amount - Principal = $2777.25 - 2000 = ₹\ 777.25$

Ex. 6 What sum of money at compound interest will amount to ₹4499.04 in 3 yr, if the rate of interest is 3% for the 1st yr, 4% for the 2nd yr and 5% for the 3rd yr?

Sol. Given that, $A = ₹\ 4499.04,\ R_1 = 3\%,\ R_2 = 4\%,\ R_3 = 5\%$ and P =?

According to the formula,

$$\text{Amount } = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right)$$

$$4499.04 = P\left(1+\frac{3}{100}\right)\left(1+\frac{4}{100}\right)\left(1+\frac{5}{100}\right)$$

$$4499.04 = P\left(\frac{103}{100}\times\frac{104}{100}\times\frac{105}{100}\right)$$

$$\therefore \quad P = \frac{4499.04\times100\times100\times100}{103\times104\times105} = ₹\,4000$$

14.1 Instalments

When a borrower pays the sum in parts, then we say that he/she is paying in instalments.

$$P = \frac{x}{\left(1+\frac{R}{100}\right)} + \frac{x}{\left(1+\frac{R}{100}\right)^2} + \frac{x}{\left(1+\frac{R}{100}\right)^3} + \ldots\cdots + \frac{x}{\left(1+\frac{R}{100}\right)^n}$$

where x = Value of each instalment

Total amount paid in instalments,

$$A = P\left(1+\frac{R}{100}\right)^n$$

where n =Number of instalments.

Ex. 7 Sapna borrowed some money on compound interest and returned it in 3 yr in equal annual instalments. If rate of interest is 15% per annum and annual instalment is ₹486680, then find the sum borrowed.

Sol. Given that, rate of interest, $R = 15\%\ pa$, Annual instalment, $x = ₹\,486680$

Total number of instalments, $n = 3$

$$\therefore \quad P = \frac{x}{\left(1+\frac{R}{100}\right)} + \frac{x}{\left(1+\frac{R}{100}\right)^2} + \frac{x}{\left(1+\frac{R}{100}\right)^3}$$

$$\Longrightarrow \quad P = x\left[\frac{100}{(100+R)} + \frac{100^2}{(100+R)^2} + \frac{100^3}{(100+R)^3}\right]$$

$$= 486680\left[\frac{100}{(100+15)} + \frac{100^2}{(100+15)^2} + \frac{100^3}{(100+15)^3}\right]$$

$$= 486680\left[\frac{20}{23} + \left(\frac{20}{23}\right)^2 + \left(\frac{20}{23}\right)^3\right]$$

$$= 486680\times\frac{20}{23} + \left(1+\frac{20}{23}+\frac{400}{529}\right) = 1111200$$

$\therefore$ Principal borrowed = ₹ 1111200

14.2 Fast Track Techniques to solve the QUESTIONS

Technique 1

Difference between Compound Interest(CI) and Simple Interest(SI)

1. Difference between CI and SI for 3 yr(D)=$P\left(\frac{R}{100}\right)^2 = \frac{SI \times R}{200}$
2. Difference between CI and SI for 3 yr(D)=$P\left(\frac{R}{100}\right)^2\left(\frac{R}{100}+3\right)$

Note: *SI and CI for one year on the same sum and at same rate are equal*

Ex. 8 The difference between compound interest and simple interest for 2 yr at rate of 5% per annum is ₹5, then find the sum.

Sol. Given that, difference(D) = ₹ 5, rate(R) = 5%

Then, according to the formula,

$$D = P\left(\frac{R}{100}\right)^2 \quad \Longrightarrow \quad 5 = P\left(\frac{5}{100}\right)^2$$

$$\Longrightarrow \quad 5 = \frac{P \times 5 \times 5}{100 \times 100} \quad \Longrightarrow \quad P = ₹\ 2000$$

Ex. 9 The difference between CI and SI for 3 yr at the rate of 20% pa is ₹152. What is the principal lent?

Sol. Difference between CI and SI for 3 yrs = ₹ 152

$$\therefore \quad P\left(\frac{R}{100}\right)^2\left(\frac{R}{100}+3\right) = 152$$

$$\Longrightarrow \quad P\left(\frac{20}{100}\right)^2\left(\frac{20}{100}+3\right) = 152$$

$$\Longrightarrow \quad P\left(\frac{1}{25}\right)\left(\frac{16}{5}\right) = 152$$

$$\Longrightarrow \quad P = \frac{152 \times 25 \times 5}{16} = 9.5 \times 25 \times 5 = ₹\ 1187.5$$

Technique 2

If simple interest for a certain sum for 2 yr at the annual rate of interest R% is SI. Then,

$$\text{Compound Interest (CI)} = SI\left(1+\frac{R}{100}\right)$$

Note: This formula is applicable only for 2 yr.

Ex. 10 If the simple interest for a certain sum for 2 yr at 5% pa is ₹200, then what will be the compound interest for same sum for same period and at the same rate of interest?

Sol. Given that, $SI = ₹\,200,$ and $R = 5\%$

According to the formula,

$$CI = SI\left(1 + \frac{R}{100}\right) = 200\left(1 + \frac{5}{100}\right)$$
$$= 200 \times \frac{205}{200} = ₹\,205$$

Technique 3

If a certain sum at compound interest becomes x times in n_1 yr and y times in n_2 yr,then

$$x^{\frac{1}{n_1}} = y^{\frac{1}{n_2}}$$

Ex. 11 If a certain sum at compound interest becomes double in 5 yr, then in how many years, it will be 16 times at the same rate of interest?

Sol. If sum=x, then

x becomes $2x$ in 5 yr.
x becomes $2x$ in 10 yr.
x becomes $4x$ in 15 yr.
x becomes $8x$ in 20 yr.

Fast Track Method

Here, $n_1 = 5\ yr,\ x = 2,\ y = 16$ and $n_2 =$?

According to the formula,

$$x^{\frac{1}{n_1}} = y^{\frac{1}{n_2}} \implies 2^{\frac{1}{5}} = 16^{\frac{1}{n_2}}$$
$$\implies \quad 2^{\frac{1}{5}} = (2)^{4\times\frac{1}{n_2}}$$
$$\implies \quad \frac{1}{5} = \frac{4}{n_2} \qquad \text{[on comparing both sides]}$$
$$\therefore \quad n_2 = 5 \times 4 = 20\ yr$$

Technique 4

If a certain sum at compound interest becomes A_1 in n yr and A_2 in $(n+1)$ yr, then

(i) Rate of compound interest $= \dfrac{A_2 - A_1}{A_1} \times 100\%$

(ii) Sum$= A_1 \left(\dfrac{A_1}{A_2}\right)^n$

Ex. 12 A sum of money invested at compound interest amounts to ₹800 in 2 yr and ₹840 in 3 yr. Find the rate of interest per annum and the sum.

Sol. SI on ₹ 800 for 1 yr $= 840 - 800 =$ ₹ 40

According to the formula,

$$SI = \frac{P \times R \times T}{100}$$

$$\Longrightarrow \quad 40 = \frac{800 \times R \times 1}{100}$$

$$\Longrightarrow \quad R = 5\%$$

Now, let the sum $= x$, then

$$\text{Amount} = \text{Sum}\left(1 + \frac{R}{100}\right)^T$$

$$x\left(1 + \frac{5}{100}\right)^2 = 800$$

$$\Longrightarrow \quad \frac{105}{100} \times \frac{105}{100} \times x = 800$$

$$\therefore \quad x = \frac{800 \times 100 \times 100}{105 \times 105} = \frac{320000}{441} = ₹\ 725.62$$

Fast Track Method

Here, $A_1 =$ ₹ 800 and $A_2 =$ ₹ 840

According to the formula,

$$\text{Rate of compound interest} = \frac{A_2 - A_1}{A_1} \times 100\% = \frac{840 - 800}{800} \times 100\% = 5\%$$

$$\text{and sum} = A_1\left(\frac{A_1}{A_2}\right)^n = 800 \times \left(\frac{800}{840}\right)^2$$

$$= 800 \times \frac{800}{840} \times \frac{800}{840} = ₹\ 725.62$$

Technique 5

If the population of a city is P and it increases with the rate of R% per annum, then

$$\text{Population after } n\ yr = P\left(1+\frac{R}{100}\right)^n$$

$$\text{Population } n\ yr \text{ ago} = \frac{P}{\left(1+\frac{R}{100}\right)^n}$$

Note

- If population decreases with the rate of R%, then (-) sign will be used in place of (+) in the above mentioned formula.
- If the rate of growth per year is R_1%, R_2%, R_3%, $\ldots R_n$%, then

$$\text{Population after } n\ yr = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right)\ldots\left(1+\frac{R_n}{100}\right)$$

(This formula can also be used, if there is increase/decrease in the price of an article.)

Ex. 13 The population of a city increases at the rate of 15% pa. If its population was 4000 at the end of year 2013, then what will be its population at the end of year 2015?

Sol. Given that, $R = 15\%, n = 2\ yr$ and $P = 4000$

According to the formula,

$$\text{Population after } n\ yr = P\left(1+\frac{R}{100}\right)^n$$

$\therefore$ $$\text{Population after 2 yr} = 4000\left(1+\frac{15}{100}\right)^2$$

$$= 400 \times \frac{23}{20} \times \frac{23}{20} = 10 \times 529 = 5290$$

Ex. 14 The population of a city increases at the rate of 5% pa. If the present population of the city is 370440, then what was its population 3 yr ago?

Sol. Given that, $R = 5\%, n = 3\ yr$ and $P = 370440$

According to the formula,

$$\text{Population } n\ yr \text{ ago} = \frac{P}{\left(1+\frac{R}{100}\right)^n}$$

$$\therefore \quad \text{Population 3 } yr \text{ ago} = \frac{370440}{\left(1+\frac{5}{100}\right)^3} = \frac{370440}{\left(\frac{21}{20}\right)^3}$$
$$= \frac{370440 \times 8000}{9261} = 40 \times 8000 = 320000$$

Ex. 15 The population of a particular area A of a city is 5000. It increases by 10% in 1st yr. It decreases by 20% in the 2nd yr because of some reason. In the 3rd yr, the population increases by 30%. What will be the population of area A at the end of 3 yr?

Sol. Given that, $P = 5000\%$, $R_1 = 10\%$, $R_1 = -20\%$ (decrease), and ($R_3 = 10\%$

$$\text{Population at the end of 3rd year} = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right)$$
$$= 5000\left(1+\frac{10}{100}\right)\left(1-\frac{20}{100}\right)\left(1+\frac{30}{100}\right)$$
$$= 5000\frac{11}{10} \times \frac{4}{5} \times \frac{13}{10}$$
$$= 10 \times 11 \times 4 \times 13 = 5720$$

Ex. 16 The cost price of a car is ₹400000. If its price decreases 10% every year, then what will be the cost of car after 3 yr?

Sol. Given that, $P = ₹\ 400000$, $R = -10\%$, and $n = 3$

According to the formula,

$$\text{Price of car after } n \text{ } yr = P\left(1+\frac{R}{100}\right)^n$$

$$\text{Price of car after 3 yr} = 400000\left(1-\frac{10}{100}\right)^3 \qquad \text{[- sign because of the decrease in price]}$$
$$= 40000 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10}$$
$$= 400 \times 9 \times 9 \times 9 = ₹\ 291600$$